VistaLux Entertainment: 2025 Growth Outlook and Modeling Methodology

Callie Mueller, Micah Pew, Aidan Christensen

```
library(vroom)
library(tidyverse)
library(forecast)
library(patchwork)
library(lmtest)
library(prophet)

predictors <- c("Income", "Production", "Savings", "Unemployment")

dir.create("figures", showWarnings = FALSE)

data <- vroom("data/CompanyGrowth.csv") %>%
  mutate(Qrtr = factor(Qrtr))

data_all <- data %>%
  filter(!is.na(PctGrowth), if_all(all_of(predictors), ~ !is.na(.x))) %>%
  mutate(q  = as.numeric(as.character(Qrtr)),
         ds = as.Date(paste(Year, (q - 1) * 3 + 1, 1, sep = "-")),
         y  = PctGrowth) %>%
  arrange(ds)

tail(data_all)
```

```
# A tibble: 6 × 10
  PctGrowth Income Production Savings Unemployment  Year Qrtr      q ds
      <dbl>  <dbl>      <dbl>   <dbl>        <dbl> <dbl> <fct> <dbl> <date>
1     0.417  1.67       0.564 17.4          -0.1    2024 1         1 2024-01-01
2     0.983  0.662      1.12  -2.72          0      2024 2         2 2024-04-01
3     0.853  0.806      1.26  -0.0857       -0.3    2024 3         3 2024-07-01
4     0.357  0.695      0.948  5.03          0.200  2024 4         4 2024-10-01
5     0.283  1.10      -0.488  9.76         -0.1    2025 1         1 2025-01-01
6     1.11   0.593     -0.540 -4.26         -0.1    2025 2         2 2025-04-01
# ℹ 1 more variable: y <dbl>
```

```r
make_future <- function(train, end_date = as.Date("2025-10-01")) {
  tibble(ds = seq(max(train$ds), end_date, by = "quarter")[-1]) %>%
    mutate(Year = year(ds),
           Qrtr = factor(quarter(ds), levels = levels(train$Qrtr)))
}

future_all <- make_future(data_all)
future_all   # CHECK: these are the quarters being projected
```

```
# A tibble: 2 × 3
  ds          Year Qrtr
  <date>     <dbl> <fct>
1 2025-07-01  2025 3
2 2025-10-01  2025 4
```

```r
ts_of <- function(x, df) ts(x, start = c(df$Year[1], df$q[1]), frequency = 4)
mae   <- function(actual, pred) mean(abs(actual - pred))
```

```r
forecast_growth <- function(train, test) {
  X <- test
  for (v in predictors) {
    f <- as.formula(paste(v, "~ Year + Qrtr"))
    X[[v]] <- predict(lm(f, data = train), newdata = test)
  }
  growth_fit <- lm(PctGrowth ~ Income + Production + Savings + Unemployment,
                   data = train)
  predict(growth_fit, newdata = X)
}

holdout_results <- function(cut_year) {
  tr <- data_all %>% filter(Year < cut_year)
  te <- data_all %>% filter(Year >= cut_year)

  known_fit <- lm(PctGrowth ~ Income + Production + Savings + Unemployment,
                  data = tr)

  tibble(
    holdout         = paste0(cut_year, "-2024"),
    n_quarters      = nrow(te),
    mae_model       = mae(te$PctGrowth, forecast_growth(tr, te)),
    mae_if_X_known  = mae(te$PctGrowth, predict(known_fit, newdata = te)),
```

```
    mae_mean_only   = mae(te$PctGrowth, mean(tr$PctGrowth))
  )
}

bind_rows(holdout_results(2024), holdout_results(2020)) %>%
  mutate(across(starts_with("mae"), ~ round(.x, 3)))
```

```
# A tibble: 2 × 5
  holdout   n_quarters mae_model mae_if_X_known mae_mean_only
  <chr>          <int>     <dbl>          <dbl>         <dbl>
1 2024-2024          6     0.28           0.165         0.316
2 2020-2024         22     0.231          0.102         0.226
```

```
current_model <- lm(PctGrowth ~ Income + Production + Savings + Unemployment,
                    data = data_all)
summary(current_model)
```

```

Call:
lm(formula = PctGrowth ~ Income + Production + Savings + Unemployment,
    data = data_all)

Residuals:
     Min       1Q   Median       3Q      Max
-0.90555 -0.15821 -0.03608  0.13618  1.15471

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.253105   0.034470   7.343 5.71e-12 ***
Income        0.740583   0.040115  18.461  < 2e-16 ***
Production    0.047173   0.023142   2.038   0.0429 *
Savings      -0.052890   0.002924 -18.088  < 2e-16 ***
Unemployment -0.174685   0.095511  -1.829   0.0689 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3102 on 193 degrees of freedom
Multiple R-squared:  0.7683,    Adjusted R-squared:  0.7635
F-statistic:   160 on 4 and 193 DF,  p-value: < 2.2e-16
```

```r
data_all <- data_all %>%
  mutate(fitted_vals = fitted(current_model),
         resids = residuals(current_model),
         time = Year + (as.numeric(as.character(Qrtr)) - 1) / 4)

title_theme <- theme(plot.title = element_text(size = 10))

p_acf <- forecast::ggAcf(data_all$resids) +
  labs(title = "ACF of MLR Residuals") + title_theme

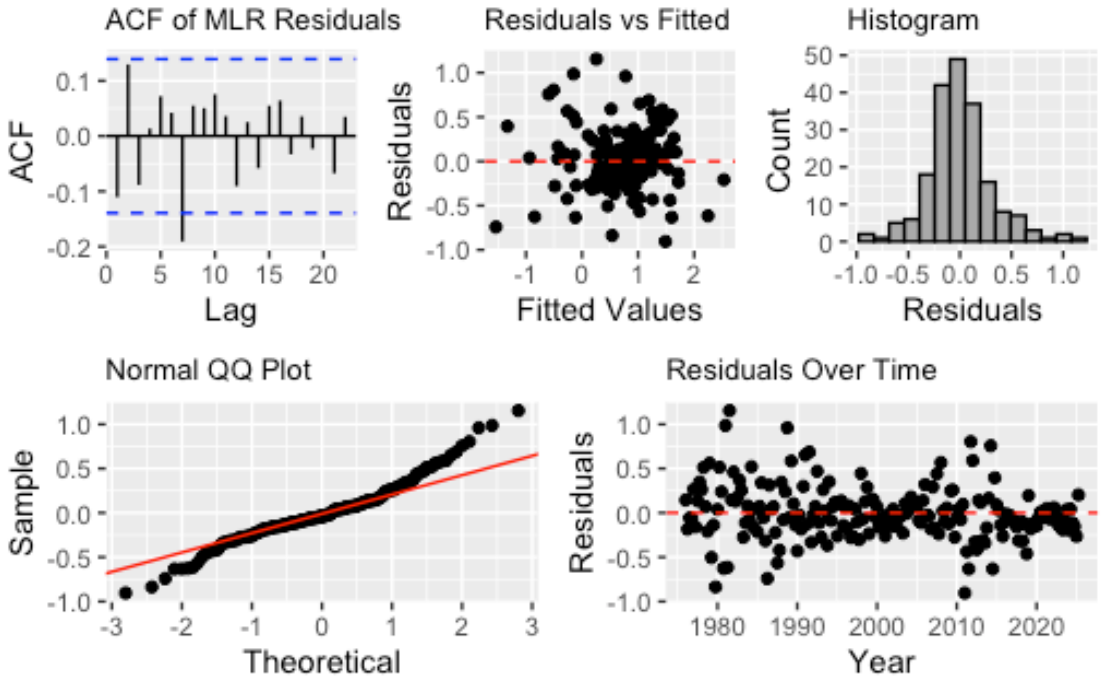
p_res <- ggplot(data_all, aes(x = fitted_vals, y = resids)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Residuals vs Fitted", x = "Fitted Values", y = "Residuals") +
  title_theme

p_hist <- ggplot(data_all, aes(x = resids)) +
  geom_histogram(bins = 15, fill = "darkgray", color = "black") +
  labs(title = "Histogram", x = "Residuals", y = "Count") + title_theme

p_qq <- ggplot(data_all, aes(sample = resids)) +
  stat_qq() + stat_qq_line(color = "red") +
  labs(title = "Normal QQ Plot", x = "Theoretical", y = "Sample") + title_theme

p_time <- ggplot(data_all, aes(x = time, y = resids)) +
  geom_point() +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  labs(title = "Residuals Over Time", x = "Year", y = "Residuals") + title_theme

(p_acf | p_res | p_hist) / (p_qq | p_time)
```




```
print(shapiro.test(data_all$resids))
```

```
    Shapiro-Wilk normality test

data:  data_all$resids
W = 0.96039, p-value = 2.397e-05
```

```
print(bptest(current_model))
```

```
    studentized Breusch-Pagan test

data:  current_model
BP = 11.193, df = 4, p-value = 0.02448
```

```
print(dwtest(current_model))
```

```
    Durbin-Watson test

data:  current_model
DW = 2.218, p-value = 0.9333
alternative hypothesis: true autocorrelation is greater than 0
```

```
print(Box.test(data_all$resids, lag = 8, type = "Ljung-Box"))
```

```
    Box-Ljung test

data:  data_all$resids
X-squared = 17.108, df = 8, p-value = 0.029
```

```
lm_income     <- lm(Income ~ Year + Qrtr,       data = data_all)
lm_production <- lm(Production ~ Year + Qrtr,   data = data_all)
lm_savings    <- lm(Savings ~ Year + Qrtr,      data = data_all)
lm_unemp      <- lm(Unemployment ~ Year + Qrtr, data = data_all)

# Predict using future_all (which holds only Q3 and Q4)
data_2025 <- future_all %>%
  mutate(
    Income       = predict(lm_income, newdata = future_all),
    Production   = predict(lm_production, newdata = future_all),
    Savings      = predict(lm_savings, newdata = future_all),
    Unemployment = predict(lm_unemp, newdata = future_all)
  )

predictions_2025 <- predict(current_model, newdata = data_2025,
                            interval = "prediction")

data_2025 <- data_2025 %>%
  mutate(Predicted_PctGrowth = predictions_2025[, "fit"],
         Lower_PI_old = predictions_2025[, "lwr"],
         Upper_PI_old = predictions_2025[, "upr"])

# Simulation interval that includes uncertainty in the forecasted predictors
pred_models <- list(Income = lm_income, Production = lm_production,
                    Savings = lm_savings, Unemployment = lm_unemp)
pred_resids <- sapply(pred_models, residuals)
```

```r
beta <- coef(current_model)
growth_resids <- residuals(current_model)

set.seed(469)
B <- 5000

# Dynamically map over the remaining future quarters
quarters_to_predict <- as.numeric(as.character(data_2025$Qrtr))

sim_summary <- map_dfr(1:nrow(data_2025), function(i) {
  q_val <- quarters_to_predict[i]
  idx <- sample(nrow(pred_resids), B, replace = TRUE)
  base <- unlist(data_2025[i, predictors])

  Xq <- sweep(pred_resids[idx, predictors], 2, base, "+")
  sims <- as.vector(cbind(1, Xq) %*% beta) +
    sample(growth_resids, B, replace = TRUE)

  tibble(Qrtr = q_val,
         Lower_PI_sim = quantile(sims, 0.025),
         Upper_PI_sim = quantile(sims, 0.975))
}) %>% mutate(Qrtr = factor(Qrtr, levels = levels(data_all$Qrtr)))

data_2025 <- left_join(data_2025, sim_summary, by = "Qrtr")
print(data_2025 %>%
        select(Year, Qrtr, Predicted_PctGrowth, starts_with("Lower"),
               starts_with("Upper")))
```

```
# A tibble: 2 × 7
   Year Qrtr  Predicted_PctGrowth Lower_PI_old Lower_PI_sim Upper_PI_old
  <dbl> <fct>               <dbl>        <dbl>        <dbl>        <dbl>
1  2025 3                   0.768       0.154        -0.781         1.38
2  2025 4                   0.657       0.0438       -0.838         1.27
# ℹ 1 more variable: Upper_PI_sim <dbl>
```

```r
#sarima

sarima_forecast_X <- function(train, h) {
  X <- sapply(predictors, function(v) {
    as.numeric(forecast(auto.arima(ts_of(train[[v]], train), seasonal = TRUE),
h = h)$mean)
  })
```

```
  matrix(X, nrow = h, dimnames = list(NULL, predictors))
}

sarima_forecast <- function(train, future) {
  h        <- nrow(future)
  X_future <- sarima_forecast_X(train, h)
  fit <- auto.arima(ts_of(train$PctGrowth, train),
                    xreg = as.matrix(train[, predictors]), seasonal = TRUE)
  fc <- forecast(fit, xreg = X_future, h = h, level = 95)

  list(fit = fit, X_future = X_future,
       out = tibble(Year = future$Year, Qrtr = future$Qrtr, Model = "SARIMA
+ xreg",
                    Predicted = as.numeric(fc$mean),
                              Lower_95   = as.numeric(fc$lower), Upper_95   =
as.numeric(fc$upper)))
}
```

```
fit_prophet <- function(df, regressors = NULL) {
  m <- prophet(weekly.seasonality = FALSE, daily.seasonality = FALSE,
               yearly.seasonality = 2, interval.width = 0.95, fit = FALSE)
  for (r in regressors) m <- add_regressor(m, r)
  suppressMessages(fit.prophet(m, df))
}

prophet_forecast_X <- function(train, future) {
  out <- map(predictors, function(v) {
    m <- fit_prophet(train %>% transmute(ds, y = .data[[v]]))
    predict(m, future %>% select(ds))$yhat
  })
  names(out) <- predictors
  as_tibble(out)
}

prophet_forecast <- function(train, future, seed = 469) {
  set.seed(seed)
  X_future <- prophet_forecast_X(train, future)
   m <- fit_prophet(train %>% select(ds, y, all_of(predictors)), regressors
= predictors)
  pred <- predict(m, bind_cols(future %>% select(ds), X_future))

  list(model = m, X_future = X_future,
```

```
        out = tibble(Year = future$Year, Qrtr = future$Qrtr, Model = "Prophet
+ regressors",
                     Predicted = pred$yhat,
                     Lower_95  = pred$yhat_lower, Upper_95  = pred$yhat_upper))
}
```

```
holdout_sarima_prophet <- function(cut_year) {
  tr <- data_all %>% filter(Year <  cut_year)
  te <- data_all %>% filter(Year >= cut_year)

  sar <- sarima_forecast(tr, te)$out$Predicted
  pro <- prophet_forecast(tr, te)$out$Predicted

  tibble(holdout = paste0(cut_year, "-", max(te$Year)),
         n_quarters = nrow(te),
         mae_sarima  = mae(te$PctGrowth, sar),
         mae_prophet = mae(te$PctGrowth, pro))
}

comparison_new           <-           bind_rows(holdout_sarima_prophet(2024),
holdout_sarima_prophet(2020)) %>%
  mutate(across(starts_with("mae"), ~ round(.x, 3)))
comparison_new
```

```
# A tibble: 2 × 4
  holdout   n_quarters mae_sarima mae_prophet
  <chr>          <int>      <dbl>       <dbl>
1 2024-2025          6      0.292       0.3
2 2020-2025         22      0.22        0.295
```

```
sarima_final  <- sarima_forecast(data_all, future_all)
prophet_final <- prophet_forecast(data_all, future_all)

proj_new <- bind_rows(sarima_final$out, prophet_final$out) %>%
  mutate(across(c(Predicted, Lower_95, Upper_95), ~ round(.x, 2)))
proj_new
```

```
# A tibble: 4 × 6
   Year Qrtr  Model                Predicted Lower_95 Upper_95
  <dbl> <fct> <chr>                    <dbl>    <dbl>    <dbl>
1  2025 3     SARIMA + xreg             0.56    -0.05     1.17
```

```
2  2025 4     SARIMA + xreg             0.74     0.13     1.34
3  2025 3     Prophet + regressors      0.67     0.07     1.26
4  2025 4     Prophet + regressors      0.49    -0.08     1.07
```

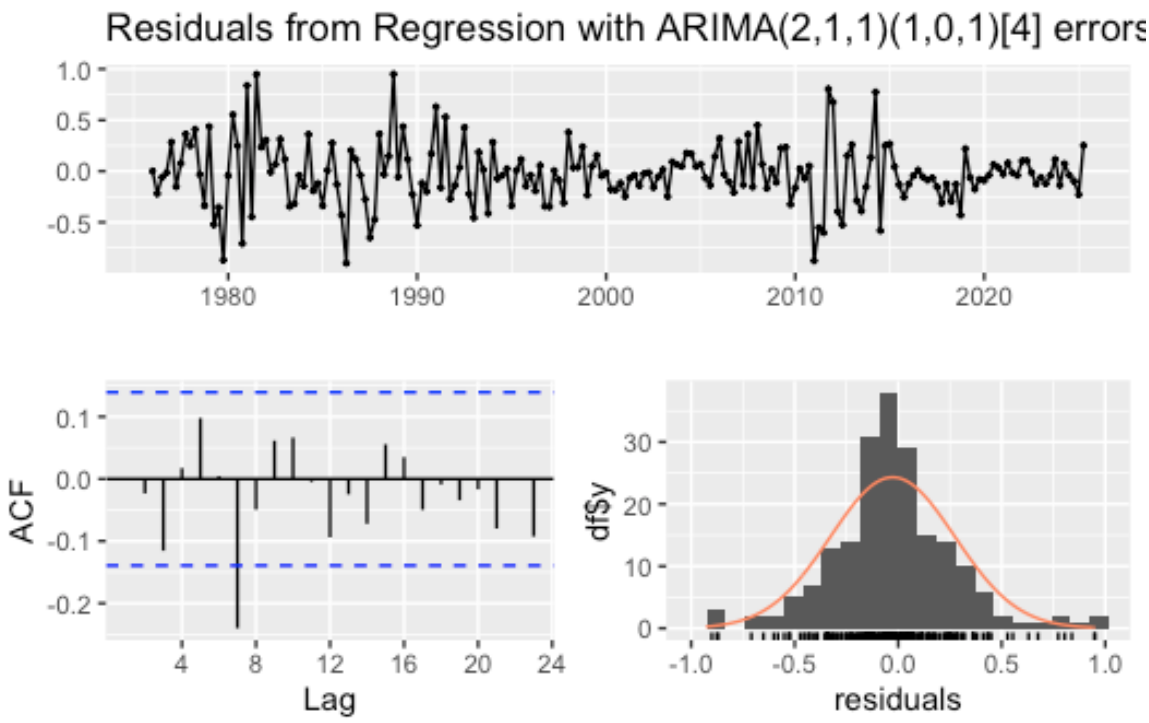
```
# Forecasted variable values each model used in stage 2 (useful for a slide)
sarima_final$X_future
```

```
        Income   Production   Savings Unemployment
[1,] 0.7950844  0.131165892 3.3408013   0.01426967
[2,] 0.7029213 -0.003253319 0.7252046  -0.05566305
```

```
prophet_final$X_future
```

```
# A tibble: 2 × 4
  Income Production Savings Unemployment
   <dbl>      <dbl>   <dbl>        <dbl>
1  0.429      0.312   -2.51      -0.136
2  0.599      0.444    2.06      -0.0748
```

```
checkresiduals(sarima_final$fit)
```




```
    Ljung-Box test

data:  Residuals from Regression with ARIMA(2,1,1)(1,0,1)[4] errors
Q* = 17.337, df = 3, p-value = 0.0006024

Model df: 5.   Total lags used: 8
```

```
sarima_resids <- as.numeric(residuals(sarima_final$fit))
shapiro.test(sarima_resids)
```

```
    Shapiro-Wilk normality test

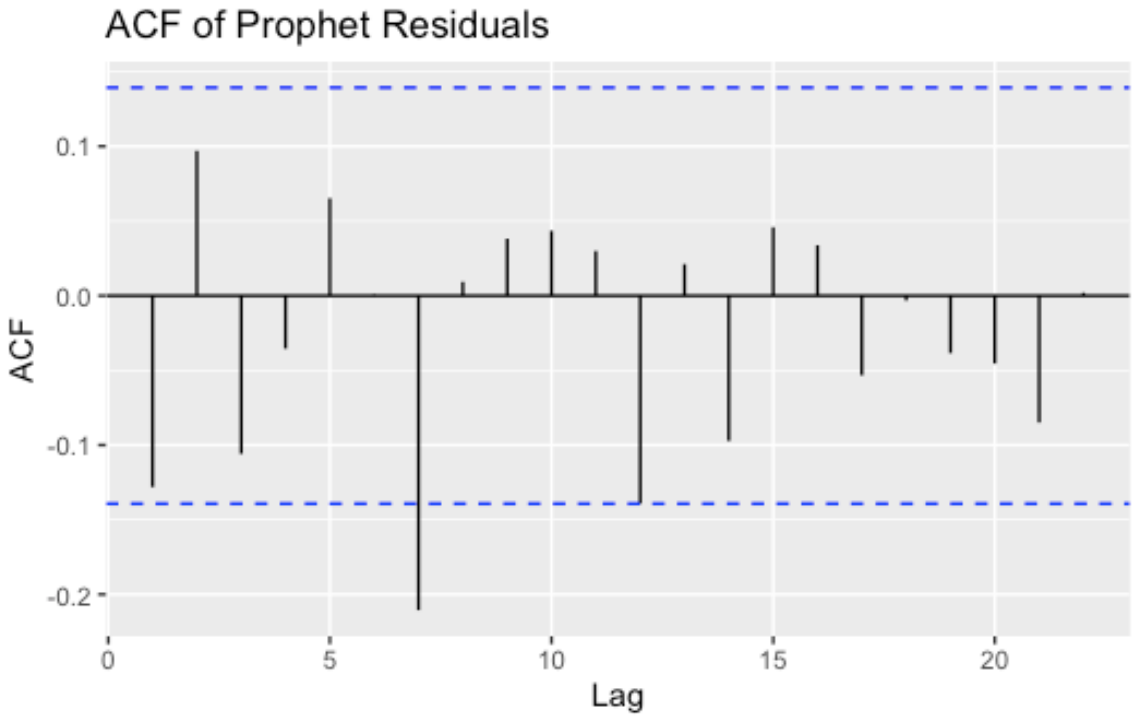
data:  sarima_resids
W = 0.9687, p-value = 0.0002126
```

```
Box.test(sarima_resids, lag = 8, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  sarima_resids
X-squared = 17.337, df = 8, p-value = 0.02678
```

```
prophet_resids <- data_all$y - predict(prophet_final$model,
                                data_all %>% select(ds, all_of(predictors)))$yhat
forecast::ggAcf(prophet_resids) + labs(title = "ACF of Prophet Residuals")
```




```
shapiro.test(prophet_resids)
```

```
    Shapiro-Wilk normality test

data:  prophet_resids
W = 0.9624, p-value = 3.985e-05
```

```
Box.test(prophet_resids, lag = 8, type = "Ljung-Box")
```

```
    Box-Ljung test

data:  prophet_resids
X-squared = 17.823, df = 8, p-value = 0.0226
```

```r
# Extract the four predictor coefficients and their standard errors
coefficient_table <- bind_rows(
  tibble(
    predictor = predictors,
    model = "MLR",
    estimate = as.numeric(coef(current_model)[predictors]),
    se = sqrt(diag(vcov(current_model)))[predictors]
  ),
  tibble(
    predictor = predictors,
    model = "SARIMA",
    estimate = as.numeric(coef(sarima_final$fit)[predictors]),
    se = sqrt(diag(sarima_final$fit$var.coef))[predictors]
  )
)

# Vary each predictor over its observed range
relationship_data <- map_dfr(predictors, function(variable) {

  values <- seq(
    min(data_all[[variable]]),
    max(data_all[[variable]]),
    length.out = 100
  )

  reference <- median(data_all[[variable]])

  crossing(
    coefficient_table |> filter(predictor == variable),
    value = values
  ) |>
    mutate(
      change = estimate * (value - reference),

      # Approximate 95% confidence intervals for the change
      margin = 1.96 * se * abs(value - reference),
```

```
      lower = change - margin,
      upper = change + margin
    )
})

# Plot the four relationships for both models
ggplot(
  relationship_data,
  aes(x = value, y = change, color = model, fill = model)
) +
  geom_hline(
    yintercept = 0,
    linetype = "dashed",
    color = "gray60"
  ) +
  geom_ribbon(
    aes(ymin = lower, ymax = upper),
    alpha = 0.15,
    color = NA
  ) +
  geom_line(aes(linewidth = model)) +
    scale_linewidth_manual(
    values = c("MLR" = 1.5, "SARIMA" = 1),
    guide = "none"
  ) +
  facet_wrap(~ predictor, scales = "free_x") +
  labs(
    title = "Predictor Relationships with Company Growth",
    subtitle = "Change relative to each predictor's median",
    x = "Predictor value (original units)",
    y = "Predicted change in PctGrowth",
    color = "Model",
    fill = "Model"
  ) +
  theme_minimal()
```



Choose the model you want to present: "SARIMA" or "MLR"

```
preferred_model <- "MLR"

variable_labels <- c(
  Income = "the disposable-income growth indicator",
  Production = "the US production growth indicator",
  Savings = "the personal savings change indicator",
  Unemployment = "the unemployment-rate change indicator"
)

board_results <- coefficient_table |>
  filter(model == preferred_model) |>
  mutate(
    lower = estimate - 1.96 * se,
    upper = estimate + 1.96 * se,

    statement = paste0(
      "A one-unit increase in ",
      variable_labels[predictor],
      " is associated with ",
      sprintf("%.3f", abs(estimate)),
```

```
      " percentage points ",
      if_else(estimate >= 0, "higher", "lower"),
      " predicted quarterly company growth, ",
      "holding the other predictors fixed."
    )
  )

# Coefficients and approximate 95% confidence intervals
board_results |>
  select(predictor, estimate, lower, upper)
```

```
# A tibble: 4 × 4
  predictor    estimate    lower   upper
  <chr>           <dbl>    <dbl>   <dbl>
1 Income         0.741   0.662    0.819
2 Production     0.0472  0.00181  0.0925
3 Savings       -0.0529 -0.0586  -0.0472
4 Unemployment  -0.175  -0.362    0.0125
```

```
# Statements your team lead can use
cat(paste(board_results$statement, collapse = "\n\n"))
```

```
A one-unit increase in the disposable-income growth indicator is associated with 0.741 percentage points higher predicted quarterly company growth, holding the other predictors fixed.

A one-unit increase in the US production growth indicator is associated with 0.047 percentage points higher predicted quarterly company growth, holding the other predictors fixed.

A one-unit increase in the personal savings change indicator is associated with 0.053 percentage points lower predicted quarterly company growth, holding the other predictors fixed.

A one-unit increase in the unemployment-rate change indicator is associated with 0.175 percentage points lower predicted quarterly company growth, holding the other predictors fixed.
```

```
# Keep observations in chronological order
evaluation_data <- data_all |> arrange(ds)
```

```r
# Start after 40 quarters of training data.
# Stop where two future observations are still available.
cutoffs <- seq(40, nrow(evaluation_data) - 2)

forecast_errors <- map_dfr(cutoffs, function(i) {

  # Only observations available at this historical cutoff
  train <- evaluation_data[1:i, ]

  # The next two quarters
  test <- evaluation_data[(i + 1):(i + 2), ]

  # Both functions forecast the economic predictors themselves
  mlr_predictions <- forecast_growth(train, test)
  sarima_predictions <- sarima_forecast(train, test)$out$Predicted

  tibble(
    cutoff = evaluation_data$ds[i],
    horizon = 1:2,
    actual = test$PctGrowth,
    mlr_error = actual - as.numeric(mlr_predictions),
    sarima_error = actual - as.numeric(sarima_predictions)
  )
})
```

```r
# Compare average absolute errors
forecast_errors |>
  group_by(horizon) |>
  summarise(
    MLR_MAE = mean(abs(mlr_error)),
    SARIMA_MAE = mean(abs(sarima_error)),
    .groups = "drop"
  )
```

```
# A tibble: 2 × 3
  horizon MLR_MAE SARIMA_MAE
    <int>   <dbl>      <dbl>
1       1   0.439      0.480
2       2   0.438      0.462
```

```r
# One-quarter-ahead comparison
errors_h1 <- forecast_errors |> filter(horizon == 1)

dm.test(
  errors_h1$mlr_error,
  errors_h1$sarima_error,
  alternative = "two.sided",
  h = 1,
  power = 1,
  varestimator = "bartlett"
)
```

```
    Diebold-Mariano Test

data:  errors_h1$mlr_errorerrors_h1$sarima_error
DM = -1.5958, Forecast horizon = 1, Loss function power = 1, p-value =
0.1126
alternative hypothesis: two.sided
```

```r
# Two-quarter-ahead comparison
errors_h2 <- forecast_errors |> filter(horizon == 2)

dm.test(
  errors_h2$mlr_error,
  errors_h2$sarima_error,
  alternative = "two.sided",
  h = 2,
  power = 1,
  varestimator = "bartlett"
)
```

```
    Diebold-Mariano Test

data:  errors_h2$mlr_errorerrors_h2$sarima_error
DM = -1.2271, Forecast horizon = 2, Loss function power = 1, p-value =
0.2216
alternative hypothesis: two.sided
```

BOARD PRESENTATION NOTES: MODEL CHOICE

Main message: We selected multiple linear regression (MLR) to explain how the four economic indicators relate to VistaLux's quarterly growth. It is the simpler of the two models and provides the clearest interpretation for the board.

Evidence supporting the decision: Our historical forecasting comparisons did not find sufficient evidence that MLR and SARIMA differ in average absolute forecast error, either one quarter or two quarters ahead. The corresponding test p-values were approximately 0.113 and 0.222. MLR had slightly lower average absolute errors in these comparisons, but the differences were not statistically significant.

This does not prove the models are equally accurate. It means these comparisons did not establish a clear forecasting advantage for either method. The tests provide supporting evidence rather than a guarantee about future performance.

Why choose MLR? Without a demonstrated forecasting advantage from SARIMA's added complexity, we favor MLR's simpler structure. Its coefficients let us explain each economic indicator's relationship with company growth while holding the other indicators fixed.

Suggested presenter wording: "We compared a straightforward regression model with a more complex time-series model. Our historical tests did not establish a clear difference in their forecasting accuracy over the next one or two quarters. We therefore selected the simpler regression model, which makes the economic relationships easier to explain and communicate to the board."

How to present the four relationships: Use the MLR estimates and uncertainty intervals from board_results. Explain the direction and size of each association. Do not claim a clear positive or negative relationship when its interval includes zero. Check the residual diagnostics before relying on the ordinary MLR uncertainty intervals.

Income refers to growth in consumers' disposable income, not VistaLux's own income. Describe the outcome as quarterly company growth. Describe relationships as associations, not proven causes.

Scope: These forecasting comparisons concern one- and two-quarter horizons. They do not establish that MLR is best at every forecast horizon. Use observed results through 2025 Q2 and distinguish them from the forecasts for 2025 Q3 and Q4.